Communication between Jeston and the motors

Nvidia Jetson Nano



UART: send character



Signal White line:
0→backward
1→forward



DC motor: DC motor 42gp-4260 dc24v14rpm (12V/24V DC行星齿轮电机42毫米无刷电机高扭矩42GP-4260
)

UART: send character



Send serial for action(attached PDF compatible with arduino)



Servo motor: hiwonder HTD-45H

Manual control of DC motor





Robot

Character (UART)	Action
W	Forward
A	Left rotate
S	Backward
D	Right rotate
(No character send)	Stop
Spacebar	Toggle mode(Manual/ AI)

Action	Left motor (white line signal level)	Right motor (white line signal level)
Forward	0	1
Left rotate	1	1
Backward	1	0
Right rotate	0	0
Stop	-	-

Manual control: Arduino as intermediate controller on DC motor

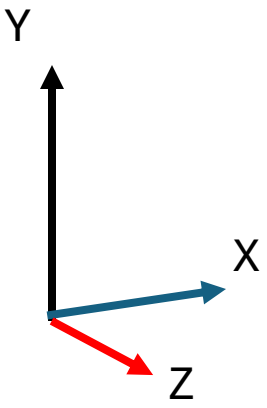


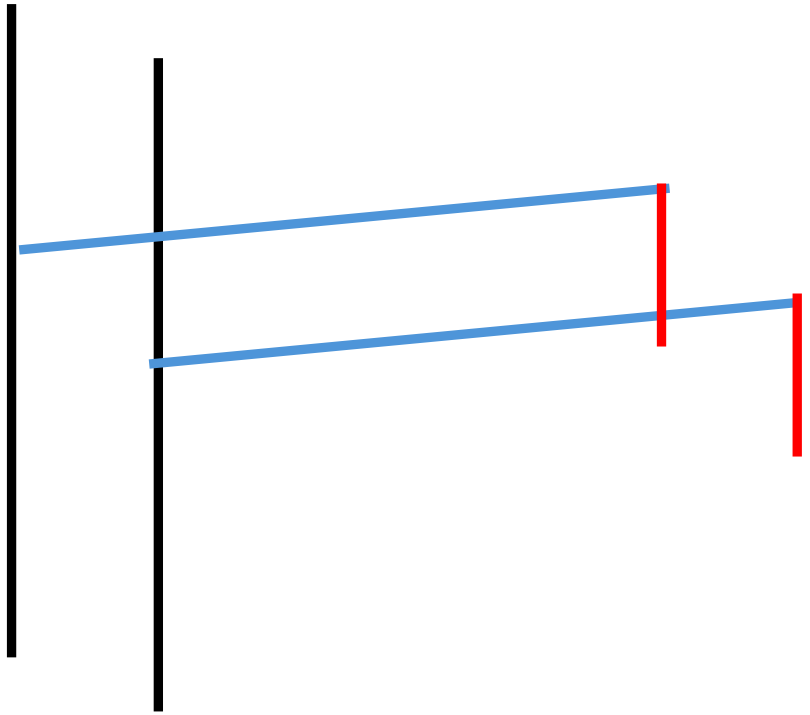


Action	Left motor (white line signal level)	Left Motor (Green Line signal level)	Right motor (white line signal level)	Left Motor (Green Line signal level)
Forward	0	1	1	1
Left rotate	1	1	1	1
Backward	1	1	0	1
Right rotate	0	1	0	1
Stop	-	0	-	0

Servo motor value ranges with coordinate X, Y, Z





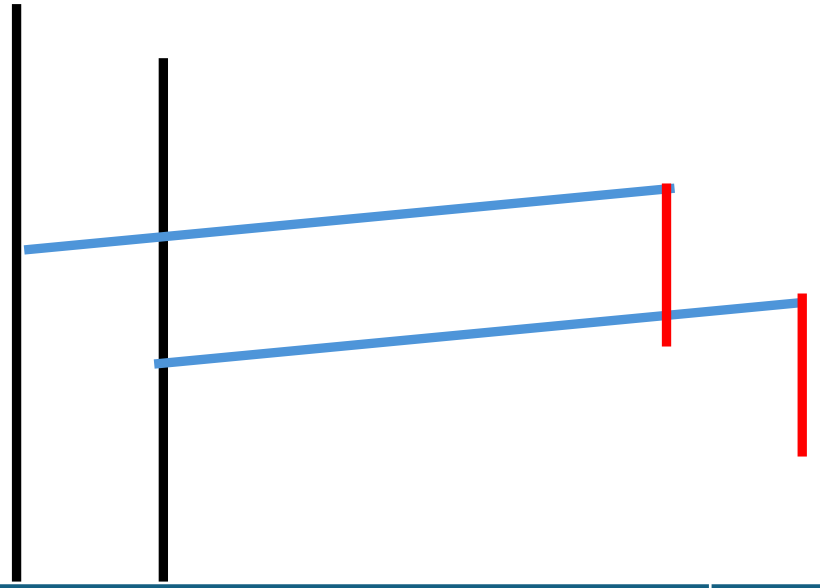


Black line: servo Y motion
Blue Line: servo X motion
Red Line: Servo Z motion(Rotate along X-axis)
(Each single line represents a servo motor)

Servo	Left (degree)	Right(degree)
Servo Y	0 to infinity(Act like DC motor)	0 to infinity(Act like DC motor)
Servo X	0 to infinity(Act like DC motor)	0 to infinity(Act like DC motor)
Servo Z	0 to 90 (Act like servo)	0 to -90 (Act like servo)

Manual mode: Servo motion with keyboard control





Black line: servo Y motion
Blue Line: servo X motion
Red Line: Servo Z motion(Rotate along X-axis)
(Each single line represents a servo motor)

Keyboard keys	Action
G	Grab(Servo X forward→ Servo Z to 90 degree/ -90 degree→ Servo X backward)
V	Release (Servo Z to 0 degree)
(up arrow)	Servo Y forward
(down arrow)	Servo Y backward
(No character send)	Stop
Spacebar	Toggle mode(Manual/ AI)

AI mode: Grab box at right distance with AI model 6 servo motor

If at 50-60 cm distance



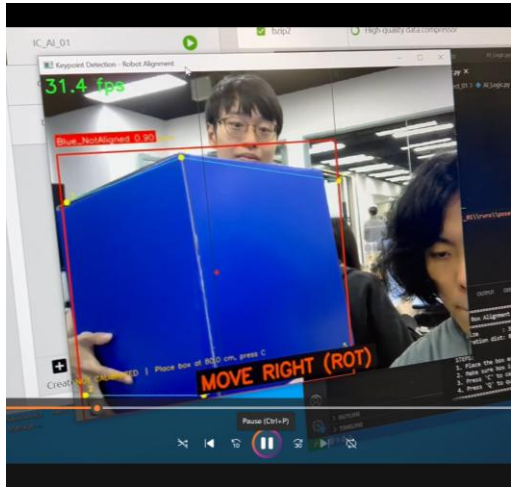


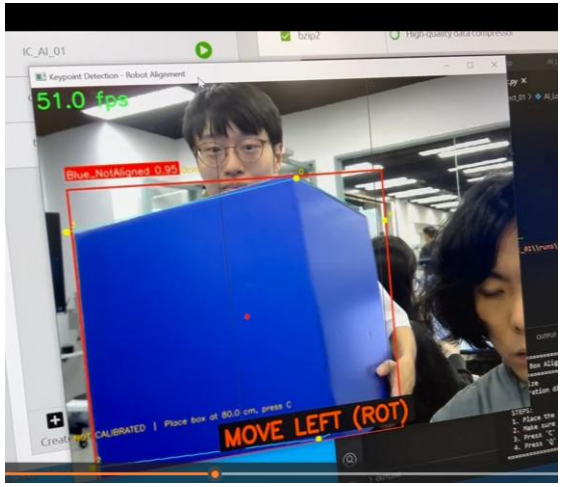
Do Action Grab → Switch to manual mode

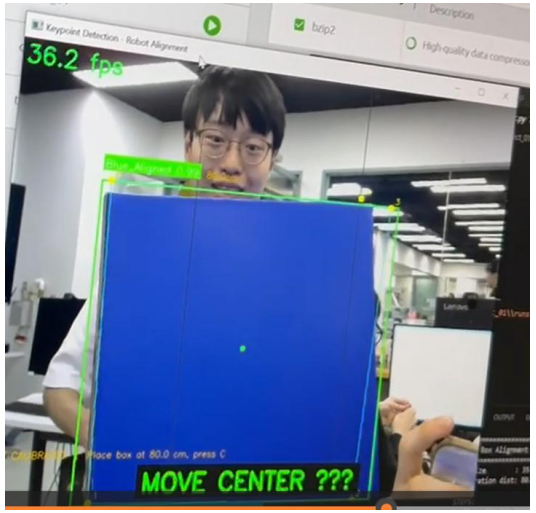
Send serial for action(attached PDF compatible with Arduino)



AI mode : Action of the DC motor (Keypoint model)

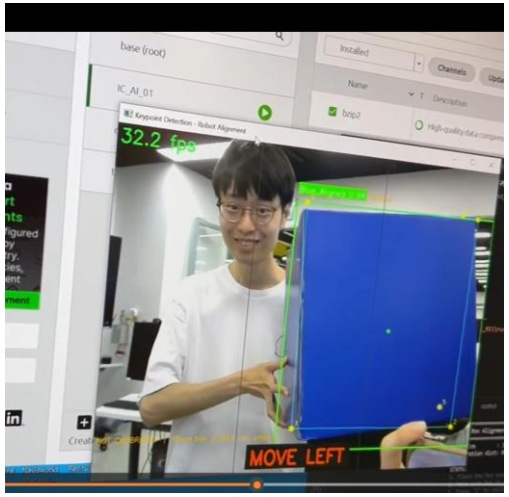


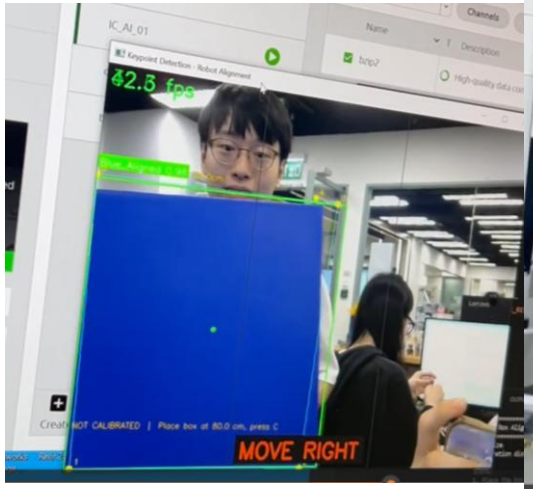


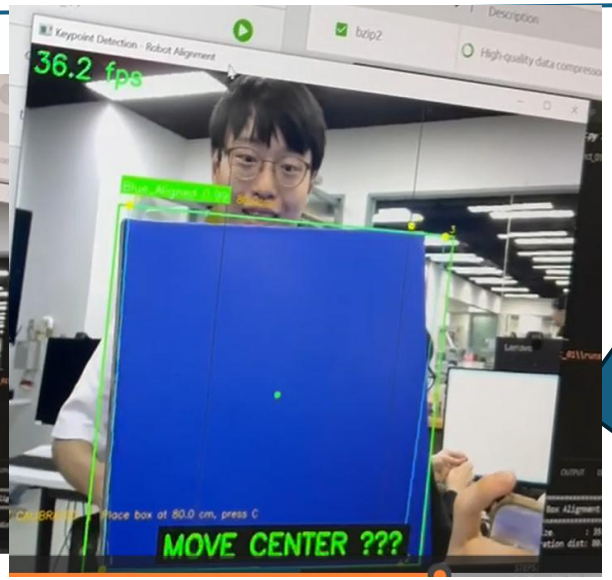


Robot minor rotation
(DC motor left/ right
moves slightly)
→(ROT) text

Robot



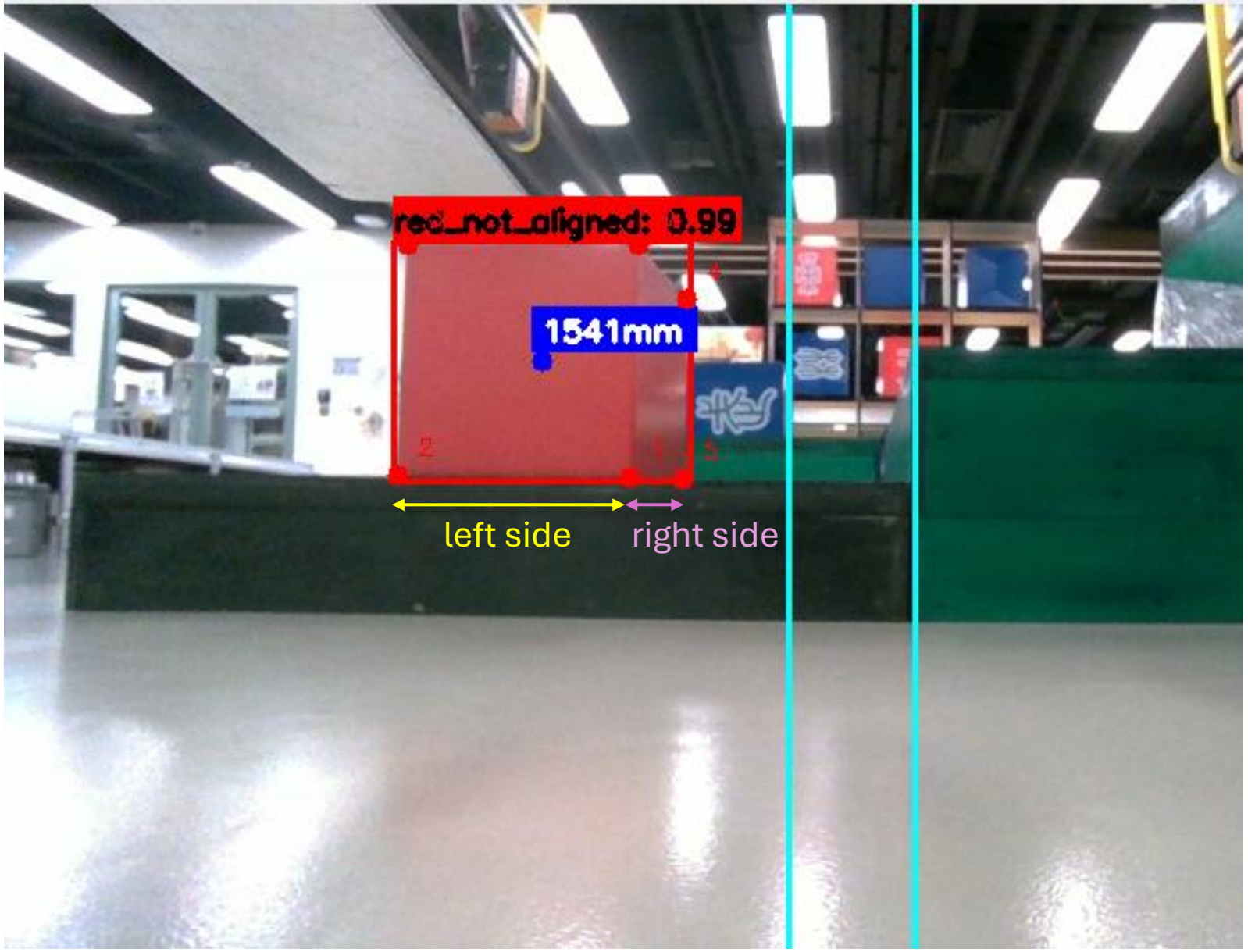




Robot navigates movement
(DC motor left/ right move
towards the box, until box in
the center of the view)
→ No (ROT) text

Robot

Movement logic using Keypoint detection model



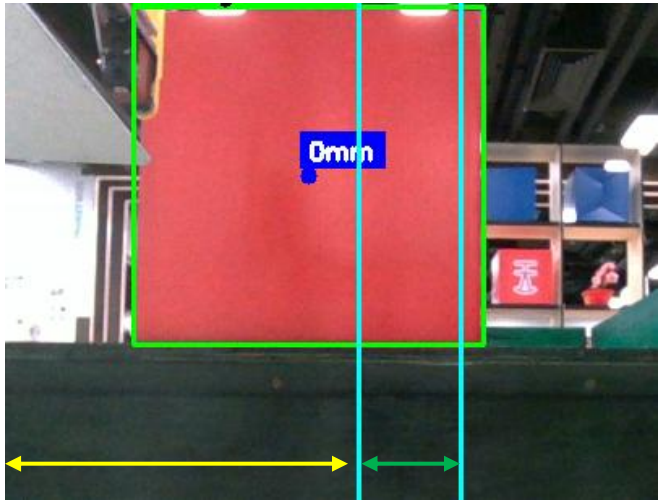
For red_not_aligned (6 keypoints):
The logic uses **keypoint distance comparison** to determine which side of the cube face is closer/further to determine rotation:

- Calculates average distance of the **left side**: (kp0→kp3 + kp1→kp2) / 2

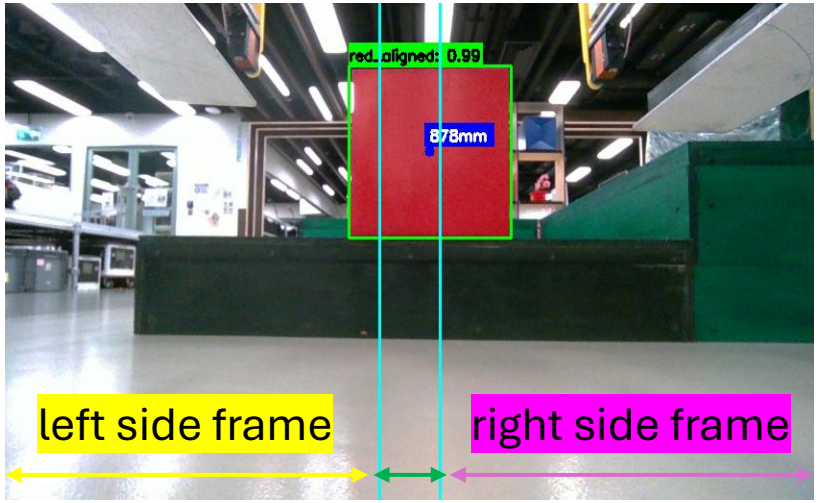
- Calculates average distance of the **right side**: (kp0→kp4 + kp1→kp5) / 2

- If left side distance < right side distance → move_right
- If left side distance > right side distance → move_left

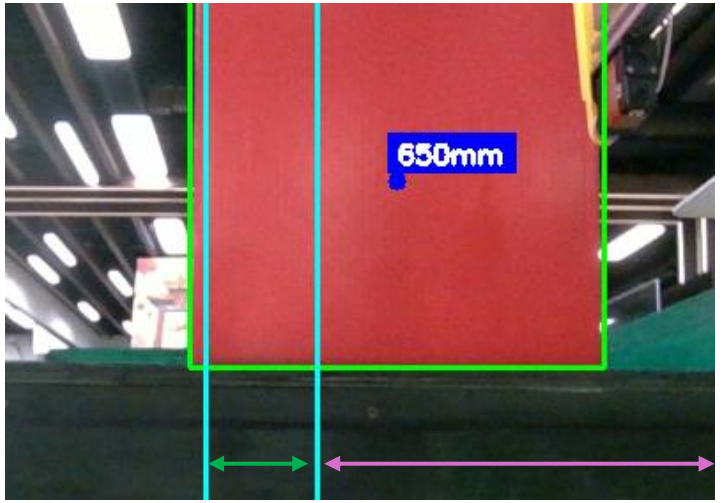
Movement logic using Keypoint detection model



left side frame center frame



center frame



center frame right side frame

For red_aligned (4 keypoints): The decision is purely based on the **bounding box center X position** relative to the frame's three sections:

Left side frame, Center frame, Right side frame

- Center X < left_boundary → move_right
- Center X in between boundaries → move_center
- Center X > right_boundary → move_left

Keypoint detection model logic

Keypoints are displayed at the closest to the bounding box

